

When does compact differentiation beat pseudospectral?

Dissipation-scale resolution, cell Reynolds number and a dealiased control in two-dimensional decaying turbulence

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Abstract

San and Staples (2012) compared high-order finite differences with a Fourier–Galerkin method in decaying two-dimensional turbulence, and organized resolution by the cell Reynolds number Re_c . We revisit that comparison with controls. Explicit sixth-order, Lele’s compact sixth-order and pseudospectral differentiation, the last undealiased, with the 2/3 rule and with 3/2 padding, share one solver and are defined by their symbols in Fourier space, so that only the differentiation changes; equivalent sparse and Fourier implementations of explicit sixth order agree to round-off, and the cost uses each scheme’s fastest implementation. Accuracy is the resolved spectral bandwidth against 2048^2 explicit sixth-order references, cross-validated against compact 2048^2 runs for the most demanding cases. Cost at equal resolved range is estimated from measured GPU step times, each scheme’s advective time-step limit, and the measured growth of the explicit scheme’s resolved range with the grid. In five decaying cases on two grids, cases with the same Re_c fall in every regime. The ranking is organized instead by how well the grid resolves the flow’s realized enstrophy-dissipation scale, $k_{\max}l_\eta$, computed a posteriori from the reference; the energy at the cutoff relative to the peak, χ , orders it approximately.

- For $k_{\max}l_\eta \lesssim 1$, where no scheme resolves the flow, explicit sixth order has the smallest bandwidth error and costs the least.
- For $1 \lesssim k_{\max}l_\eta \lesssim 2$ compact sixth order leads. Wherever $k_{\max}l_\eta \gtrsim 1$ it resolves 1.3–1.6 times the shells of explicit sixth order, at 0.44–0.81 times its estimated cost at equal range.
- For $k_{\max}l_\eta \gtrsim 2$ –2.5 pseudospectral differentiation resolves the most, up to 1.7 times the shells of explicit sixth order, at 0.54–0.76 times its cost.

These boundaries lie near the thresholds quoted for a resolved ($k_{\max}l_\eta > 1$) and a well-resolved (≥ 2) two-dimensional simulation, but rest on the same ten points that reveal the ordering. Pseudospectral differentiation piles energy up at the cutoff, and 3/2 padding leaves the pile-up and the resolved bandwidth unchanged (within 8 shells in all ten cases) at 1.85 times the step cost: with the skew-symmetric nonlinear term used throughout, the crossover is not an artifact of quadratic aliasing. The organization holds for transient decaying turbulence, not for all statistics: in a forced, statistically stationary case at $k_{\max}l_\eta = 1.33$ the schemes cannot be told apart in the inertial range within the sampling scatter of the forcing.

1 Introduction

Finite differences of high order are attractive for turbulence simulation because they combine accuracy at the resolved scales with the flexibility of local operators. Lele’s compact schemes [1] are the standard example. A tridiagonal implicit relation between derivatives and values gives a modified wavenumber close to the exact one over most of the resolved range, so they are described as having spectral-like resolution. Laizet and Lamballais [2] built on this a simple and efficient solver for incompressible flow with quasi-spectral accuracy. It used sixth-order compact schemes, a modified-wavenumber analysis of every operator, and a Poisson equation solved in spectral

space. Through a series of validation cases they showed that, at marginal resolution, the compact schemes preserve quantities sensitive to the small scales much better than second-order schemes, for a moderate extra cost. They also argued that compact differentiation of the nonlinear terms is cheaper than pseudospectral differentiation. The resulting code, Incompact3d [3, 4], scales to $O(10^5)$ cores, and its successor Xcompact3d [5] is a widely used open-source framework. The quasi-spectral philosophy is therefore well established.

The comparison with spectral methods at the resolution limit has a long history in two dimensions. Fox and Orszag [19] and Herring et al. [20] computed decaying two-dimensional turbulence spectrally; the latter found a spectral code equivalent in accuracy to a grid-point code with twice the resolution. Browning and Kreiss [21] built convergent pseudospectral reference solutions from an estimate of the smallest scale and compared common methods against them: a fourth-order scheme on the same grid was essentially identical to the reference when the energy lay at low wavenumbers, while truncated (“chopped”) and hyperviscous solutions differed considerably. For large-eddy simulation, Ghosal [6] and Kravchenko and Moin [7] showed that aliasing errors are more destructive for spectral and high-order finite-difference calculations than for low-order ones, and that the form of the nonlinear term decides whether a scheme conserves energy in the presence of aliasing. Moin and Mahesh [8] summarise the picture: finite-difference schemes carry larger differentiation errors but lower aliasing errors. Vreman, Geurts and Kuerten [22] found that the ranking of second- and fourth-order schemes reverses with the ratio of filter width to grid spacing. Fedioun, Lardjane and Gökalp [13] separated aliasing from truncation errors and found high-resolution compact differences nearly as accurate as dealiased pseudospectral differentiation in large-eddy simulation of decaying isotropic turbulence, and Park, Yoo and Choi [14] found the finite-differencing error to outweigh the aliasing error, with the skew-symmetric form reducing both.

The closest work to ours is that of San and Staples [9]. They compared explicit central differences of orders 2–6, compact schemes of orders 4 and 6, Arakawa and dispersion-relation-preserving schemes with a Fourier–Galerkin method in decaying two-dimensional turbulence, in the vorticity–streamfunction form with a fast Poisson solver shared by all schemes. Sixth-order schemes agreed with the spectral method, at speed-ups of about 5 in CPU time. They organized the resolution by the cell Reynolds number $Re_c = Re h$ ($Re 2\pi/N$ on their domain). Below $Re_c \approx 2$ all schemes agreed; for larger Re_c the order mattered; and above $Re_c \approx 30$ the non-conservative schemes piled energy up at small scales, which they attributed to aliasing and which the conservative Arakawa scheme removed. Their spectral reference was dealiased by padding to $2N$ points, and its reported cost interpolated between the times on N and $2N$ points.

Three strands of later work bear on their conclusions. *Resolution criteria.* Direct simulations are usually judged by how well the grid resolves the dissipation scale: $k_{\max}\eta$ in three dimensions [23], and in two dimensions the enstrophy-dissipation length $l_\eta = (\nu^3/\eta)^{1/6}$, with $k_{\max}l_\eta > 1$ for a resolved and ≥ 2 for a well-resolved simulation [24]. A ratio of the spectrum’s peak to its value at the cutoff serves as a pass/fail convergence diagnostic in two-dimensional pseudospectral magnetohydrodynamics [25], and the accuracy of a scheme has been measured as the bandwidth over which its spectrum follows a high-resolution reference [26]. We found no prior use of any of them to say which differentiation scheme to choose. *Pile-up at the cutoff.* The conventional view attributes it to aliasing [27]. Yet Galerkin-truncated flows thermalize at the cutoff without any aliasing [28, 16, 29], the 2/3-dealiased Fourier method loses spectral convergence like the aliased one once the solution is not smooth [30], and the pile-up near the cutoff of alias-free spectral simulations has been described as a truncation effect absent from finite differences [31]; Yalla, Oliver and Moser [15] traced a pile-up in implicitly filtered large-eddy simulation to numerical dispersion alone. We found no controlled test of the two explanations on the same marginally resolved flow. *Cost.* Spectral methods have been compared with finite differences in cost and accuracy for decades [32], and recently with effective resolution from spectra and a stability-limited time step [33]. The usable time step depends on the scheme

and on the form of the nonlinear term [34], and dealiasing strategies differ in cost at equal accuracy [35, 36]. Rodhiya, Bhattacharya and Verma [10] found forced three-dimensional turbulence computed pseudospectrally and with second-order finite differences statistically nearly the same, and Yamamoto and Tsuji [17] found a selective filter, not full 3/2 dealiasing, the most cost-effective in channel flow.

Three questions are therefore open. Is Re_c , a property of the grid and the viscosity, the right axis for choosing a scheme, or does the answer depend on the flow’s realized dissipation scale? Is the pile-up at the cutoff aliasing or truncation, tested on the same flow? And which scheme gives the most resolved bandwidth per unit of computing time, including its time step? Answering them needs four things together.

1. The schemes compared in otherwise identical solvers, so that differences come from the differentiation alone.
2. Pseudospectral differentiation as a competitor rather than as the reference: undealiased, with the 2/3 rule, and with 3/2 padding, so that aliasing can be switched off without giving up range.
3. A reference whose own discretization does not affect the measures, checked by a second reference computed with a different scheme.
4. A cost that includes the time step each scheme allows, not only the time per step, with each scheme in its fastest implementation.

This paper provides such a comparison for two-dimensional periodic turbulence, in an open code written for the purpose (cnavier). Its contributions are:

- an experimental apparatus in which explicit sixth-order, compact sixth-order and pseudospectral differentiation share one solver and are defined by their symbols in Fourier space, with sparse and Fourier implementations of the finite differences verified to agree to round-off, so that the accuracy comparison is controlled and the cost comparison can use each scheme’s fastest path (sections 2 and 3);
- the resolved bandwidth of [26] at tighter tolerances, against cross-validated references, combined with an estimate of the cost at equal resolved range that includes each scheme’s advective stability limit (section 4);
- a regime map for decaying turbulence: Re_c does not organize the ranking, while the classical two-dimensional resolution measure $k_{\max}l_\eta$ does, monotonically in our cases, with explicit sixth order ahead below about 1, compact sixth order between about 1 and 2, and pseudospectral differentiation above (section 5.1);
- a dealiased control: with the skew-symmetric nonlinear term, 3/2 padding leaves the pseudospectral pile-up and bandwidth unchanged, so quadratic aliasing of the nonlinear term does not explain the crossover (section 5.2);
- the finding that this organization does not carry over to forced, statistically stationary turbulence, where the schemes are indistinguishable in the inertial range within the sampling scatter of the forcing (section 5.4).

We do not propose a new scheme. The schemes are standard; what is new is the controlled comparison and the regime-dependent answer.

2 Numerical method

2.1 Equations and time integration

The two-dimensional incompressible Navier–Stokes equations are solved on the doubly periodic unit square in the vorticity–streamfunction form

$$\frac{\partial \omega}{\partial t} = N(\omega) + \nu \nabla^2 \omega + f, \quad \nabla^2 \psi = -\omega, \quad u = \partial_y \psi, \quad v = -\partial_x \psi, \quad (1)$$

with $\nu = 1/\text{Re}$ and f the forcing and damping terms of section 5.4. The nonlinear term takes the skew-symmetric form

$$N = -\frac{1}{2} (u D_x \omega + v D_y \omega) - \frac{1}{2} (D_x(u\omega) + D_y(v\omega)). \quad (2)$$

Because the periodic difference operators are antisymmetric, it conserves the discrete enstrophy $\frac{1}{2}\langle \omega^2 \rangle$ to round-off for every scheme here. The advective form does not, and in under-resolved turbulence creates enstrophy near the cutoff, 23–45 % of the input in the runs of the companion methodology. Time integration is the classical fourth-order Runge–Kutta method, in double precision throughout.

2.2 Operators by their symbols

On a periodic grid of n points every translation-invariant difference operator is a circulant matrix, diagonal in the discrete Fourier basis. Its eigenvalue at wavenumber k is its symbol, $\sum_j c_j e^{ikjh}$ for the stencil c_j . Writing $\tilde{D}_x(k) = ik^*(k)$ and $\tilde{D}_{xx}(k) = -(k^{**})^2$, a scheme is characterised by its modified wavenumbers. The solver reads the symbols from the first row of each one-dimensional operator and keeps them in one table. Everything that depends on the operators uses that table: the Poisson eigenvalues $\tilde{D}_{xx} + \tilde{D}_{yy}$, the energy of a mode in the spectral diagnostics, the amplitude of the random forcing, and the stability limits.

Every operator is applied in Fourier space: one forward transform of ω , then inverse transforms of the symbols times its spectrum for $D_x \omega$, $D_y \omega$, the Laplacian, the stream function and the velocity. The skew-symmetric correction takes two more forward transforms and one inverse. For a finite-difference scheme this is the same matrix as the sparse stencil, applied differently: after 20 steps of a forced run with every term active, the two paths agree to 10^{-14} . Changing the scheme changes only the symbol table. This is what makes the comparison controlled. The time integrator, the Poisson operator (always $\tilde{D}_{xx} + \tilde{D}_{yy}$ of the scheme itself), the nonlinear form, the forcing and the diagnostics are identical code for all schemes.

2.3 The schemes

Explicit sixth order (E6). The centred seven-point stencils.

Compact sixth order (C6). Lele’s tridiagonal schemes [1], $\frac{1}{3}f'_{i-1} + f'_i + \frac{1}{3}f'_{i+1} = \frac{14}{9}\delta_1 f_i + \frac{1}{9}\delta_2 f_i$ and the analogous second-derivative scheme with $\alpha = 2/11$. On a periodic grid $A^{-1}B$ is circulant; its symbols are $k^*h = (\frac{14}{9}\sin\theta + \frac{1}{18}\sin 2\theta)/(1 + \frac{2}{3}\cos\theta)$ and $(k^{**}h)^2 = -(\frac{24}{11}(\cos\theta - 1) + \frac{3}{22}(\cos 2\theta - 1))/(1 + \frac{4}{11}\cos\theta)$, $\theta = kh$.

Pseudospectral (S). The exact symbols ik and $-k^2$, with the first-derivative symbol set to zero at the Nyquist wavenumber. With exact symbols the two discrete energies $\frac{1}{2}\langle |\mathbf{u}|^2 \rangle$ and $\frac{1}{2}\langle \psi \omega \rangle$ coincide.

Pseudospectral with the 2/3 rule (S_{2/3}). Orszag’s truncation [11]: modes with $|k| \geq n/3$ along either axis are removed from every operator and from ω after each step. The nonlinear term is then a Galerkin truncation and conserves energy and enstrophy exactly in either form (10^{-14} of the largest shell transfer in our tests).

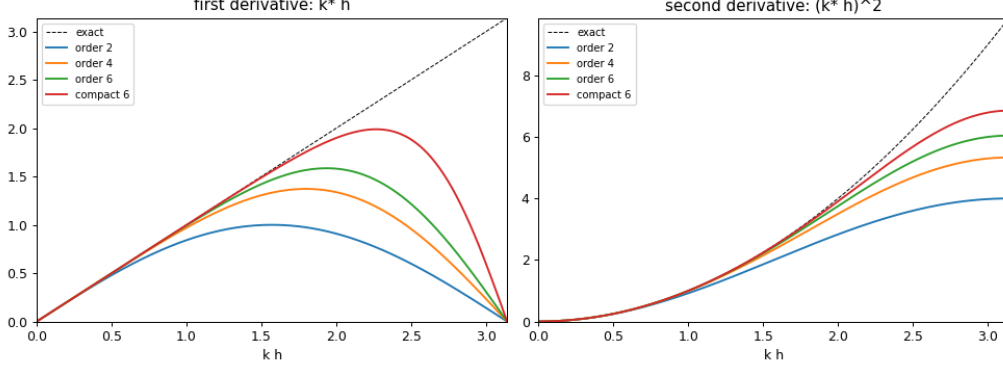


Figure 1: Modified wavenumbers of the first and second derivatives: explicit orders 2, 4 and 6, and compact sixth order. The pseudospectral symbols are the exact ones (dashed).

fraction of $[0, \pi]$ within	first derivative		second derivative		$(k^*h)_{\max}$
	1 %	10 %	1 %	10 %	
E6	0.35	0.54	0.45	0.71	1.59
C6	0.50	0.70	0.56	0.81	1.99
S	1	1	1	1	π
$S_{3/2}$	1	1	1	1	π
$S_{2/3}$	$2/3$	$2/3$	$2/3$	$2/3$	$2\pi/3$

Table 1: Fraction of the resolved range differentiated within 1 % and 10 %, and the largest modified wavenumber, which sets the advective stability limit.

Padded pseudospectral ($S_{3/2}$). The exact symbols, with the nonlinear term dealiased by padding [12]: the spectra of u , v , $D_x\omega$ and $D_y\omega$ are zero-padded to a grid 3/2 as fine, the product is formed there and its spectrum truncated back. Every mode but the Nyquist ones then carries the Fourier–Galerkin nonlinear term, free of quadratic aliasing, and no retained non-Nyquist mode is discarded. This is the dealiased pseudospectral baseline of the comparison; $S_{2/3}$ buys the same freedom from aliasing by giving up a third of the range.

Figure 1 and table 1 summarise the modified wavenumbers. The larger the resolved fraction, the larger $(k^*h)_{\max}$ and the smaller the explicit advective time step, which goes as $h/(k^*h)_{\max}$.

2.4 Spectral diagnostics

The diagnostics use the solver’s own operators. The energy of mode $\mathbf{k}$ is $\frac{1}{2}A|\hat{\psi}|^2$ with $A = |\tilde{D}_x|^2 + |\tilde{D}_y|^2$ and $\hat{\omega} = Q\hat{\psi}$, $Q = -(\tilde{D}_{xx} + \tilde{D}_{yy})$. The spectral energy and enstrophy fluxes use the solver’s nonlinear term, weighted by A/Q for the energy. With these definitions the discrete energy budget closes exactly, to round-off, rather than to the order of the scheme. The energy spectrum $E(k)$ is the shell sum of mode energies in shells of width 2π .

3 Verification

- *Order.* On a manufactured periodic solution the explicit schemes converge at orders 2, 4 and 6 and the compact scheme at 6.04, with an error at 64^2 twelve times smaller than E6. The pseudospectral error at 32^2 is 3.5×10^{-9} , against 1.1×10^{-4} for C6, and falls by 16 when the time step is halved: it is the error of the time integration.
- *Equivalence of the two operator paths.* For E6 and C6, the Fourier path reproduces the sparse operators to 10^{-14} in the fields, integrals and spectra after 20 forced steps. A decaying

turbulence run with C6 reproduces the spectrum of the sparse (banded) implementation to 4×10^{-12} .

- *Padding.* The $S_{3/2}$ nonlinear term matches the direct convolution of the modal coefficients over all non-Nyquist triads to 7×10^{-15} on 10×8 and 9×7 grids, which covers the even and odd Nyquist cases.
- *CPU and GPU.* The GPU implementation (cuFFT) matches the CPU one (FFTW) to 6×10^{-15} for all five schemes, including both dealiased ones.
- *Conservation.* The skew-symmetric form conserves enstrophy to 10^{-15} – 10^{-14} of the largest shell transfer for all schemes, and $S_{2/3}$ and $S_{3/2}$ also energy, in either form.
- *References.* Each decaying case has its own 2048² E6 reference. For the two most demanding cases ($\text{Re} = 8 \times 10^4$, $k_0 = 20$), a second 2048² reference computed with C6 agrees with it to 1.5×10^{-3} up to $K = 255$, the largest shell compared, and to 10^{-4} up to $K = 127$. The references therefore do not depend on their own discretization within the tolerances measured, 2 % and 10 %.

4 Experimental design

Decaying turbulence. The initial field has random phases, with modal amplitudes whose shell envelope is $E(k) \propto (k/k_0)^4 e^{-2(k/k_0)^2}$ peaking at $K = k_0$ ($K = |\mathbf{k}|/2\pi$), normalised to $E = \frac{1}{2}$. The phase of each wavevector depends on the seed and the wavevector only, so every grid that holds the spectrum gets the same field. All runs of a case share the time step (6.25×10^{-5}), so the spectra are compared at exactly the same instant. The comparison time keeps t/τ_0 fixed, with $\tau_0 \propto 1/k_0$ since all runs start with the same energy: $t = 0.09 \cdot 10/k_0$, about six eddy turnovers. By then the cascade has filled the spectrum, but the largest scales ($K \leq 20$) still agree with the reference to 10^{-3} – 10^{-2} , so the runs have not yet decorrelated; at $t = 0.5$ they differ by tens of per cent even at $K = 10$. The baseline is $\text{Re} = 2 \times 10^4$, $k_0 = 10$. One parameter at a time is varied: $\text{Re} = 5 \times 10^3$ and 8×10^4 , and $k_0 = 5$ and 20. Each case is run on 256^2 and 512^2 with the five schemes.

Accuracy. The resolved bandwidth is the largest shell K_{10} (K_2) up to which every shell's energy is within 10 % (2 %) of the reference. The pile-up is the largest ratio E/E_{ref} . The organizing parameter tested is

$$\chi = \frac{E_{\text{ref}}(K_{\text{Nyq}})}{\max_K E_{\text{ref}}(K)}, \quad (3)$$

the reference's energy at the grid's Nyquist shell relative to its peak, a measure of how much of the flow lies at the cutoff. It is set against the cell Reynolds number of San and Staples [9], $\text{Re}_c = \text{Re} h$ with $h = 1/n$ on the unit square. All runs start with the same energy, so Re_c is comparable across our cases, though not numerically with theirs. Re_c depends on the grid and the viscosity only; χ also depends on where the flow puts its energy, through k_0 and the time. The third axis is the classical one, $k_{\text{max}} l_\eta$, with $k_{\text{max}} = \pi/h$ the Nyquist wavenumber along an axis and $l_\eta = (\nu^3/\eta)^{1/6}$ the enstrophy-dissipation length of the reference at the comparison time, $\eta = \nu \langle |\nabla \omega|^2 \rangle$ its enstrophy dissipation (the shell sum of its viscous enstrophy dissipation, which equals 2ν times its palinstrophy to 0.1 %). η is defined as by Lunasin et al. [24]. Their k_{max} is the largest radial wavenumber the 2/3 rule keeps, $(\sqrt{2}/3)N$ in units of 2π on their unit box, which is 0.94 of ours if, as in the three-dimensional criterion $k_{\text{max}} \eta$ with $k_{\text{max}} = \pi/\Delta x$, the product is formed with angular wavenumbers. Their text does not say so explicitly; the alternative, shell units, would make their resolved runs carry some 38 grid cells per l_η , against about 6 with angular units. Like χ , $k_{\text{max}} l_\eta$ is computed here a posteriori, from the reference: it

case	grid	Re_c	χ	$k_{\max}l_\eta$	E6	C6	S	$S_{3/2}$	$S_{2/3}$
$\text{Re} = 8 \times 10^4$	256^2	313	1.2×10^{-5}	0.76	80	79	75	79	46
$k_0 = 20$	256^2	78	1.6×10^{-4}	0.92	89	74	74	72	49
baseline	256^2	78	4.9×10^{-6}	1.35	74	118	93	93	50
$\text{Re} = 8 \times 10^4$	512^2	156	2.3×10^{-7}	1.52	152	217	193	200	117
$k_0 = 20$	512^2	39	1.6×10^{-6}	1.83	159	213	217	217	130
$k_0 = 5$	256^2	78	9.0×10^{-8}	2.01	71	105	101	109	62
$\text{Re} = 5 \times 10^3$	256^2	20	2.0×10^{-7}	2.50	74	105	120	120	70
baseline	512^2	39	1.5×10^{-8}	2.71	152	198	235	239	143
$k_0 = 5$	512^2	39	4.2×10^{-11}	4.01	146	197	246	246	154
$\text{Re} = 5 \times 10^3$	512^2	9.8	2.8×10^{-12}	5.01	144	196	249	248	163

Table 2: Resolved bandwidth K_{10} (shells within 10% of the reference) in decaying turbulence, ordered by $k_{\max}l_\eta$, with the cell Reynolds number $\text{Re}_c = \text{Re}h$ and the cutoff energy χ for comparison. Nyquist shells: 127 (256^2), 255 (512^2).

classifies the runs, and choosing a scheme before running would need an a priori estimate of η , which an under-resolved run may itself misjudge.

Cost. The step time t_{step} is measured on one GPU (NVIDIA RTX 3050 laptop, double precision). At the advective limit the time step goes as $h/(k^*h)_{\max}$, so the cost per unit simulated time is proportional to $t_{\text{step}}(k^*h)_{\max}/h$. We compare at equal resolved range, which is an estimate, not a benchmark of each equal-range configuration. The resolved range of E6 grows linearly with n (74 shells on 256^2 , 152 on 512^2 at the baseline), and its cost per unit time as n^3 . Matching a scheme’s K_{10} with E6 therefore costs $(K_{10}/K_{10}^{\text{E6}})^3$ times E6 on the same grid. We report the scheme’s cost divided by that.

Accuracy and cost use each scheme’s own fastest path. The comparison isolates the discretization through the symbols, but the cost should be that of a sensible implementation. C6 and the pseudospectral variants are applied in Fourier space, which is how they are cheapest here. E6 is cheapest as a sparse seven-point stencil (table 3), so its runs and its cost use that path. The two E6 paths are the same matrix and agree to 10^{-14} (section 3), so the choice affects the cost only, and it favours E6: in Fourier space it would cost 1.4 times as much per step.

Forced turbulence. Forced runs decorrelate, so they are compared by time-mean statistics. A random forcing injects energy at rate $\varepsilon = 0.1$ in the shell $|K - 4| \leq 1$ by white-in-time kicks; a large-scale hypodrag $-\alpha_h\psi$ removes the inverse cascade. Two cases are run on 256^2 . (a) Hyperviscosity $-\nu_h(-\nabla^2)^4\omega$ confines the small-scale dissipation near the cutoff (statistics over $t = 6\text{--}16$). (b) Plain viscosity at $\text{Re} = 10^5$ with no hyperviscosity, against a 768^2 pseudospectral reference ($t = 4\text{--}10$).

5 Results

5.1 Decaying turbulence: a regime map

Table 2 and fig. 3 show the result. Re_c does not organize it. The three 256^2 cases at $\text{Re}_c = 78$, differing only in k_0 , fall in three different regimes: explicit sixth order leads at $k_0 = 20$, compact clearly at $k_0 = 10$, padded pseudospectral and compact at $k_0 = 5$. At $\text{Re}_c = 39$, pseudospectral differentiation resolves 1.68 times the shells of E6 at $k_0 = 5$ and 1.36 at $k_0 = 20$. At fixed Re and grid the initial spectrum changes the enstrophy dissipation, and with it l_η , by a factor of two; Re_c cannot see that.

Ordered by $k_{\max}l_\eta$, the leading scheme changes monotonically.

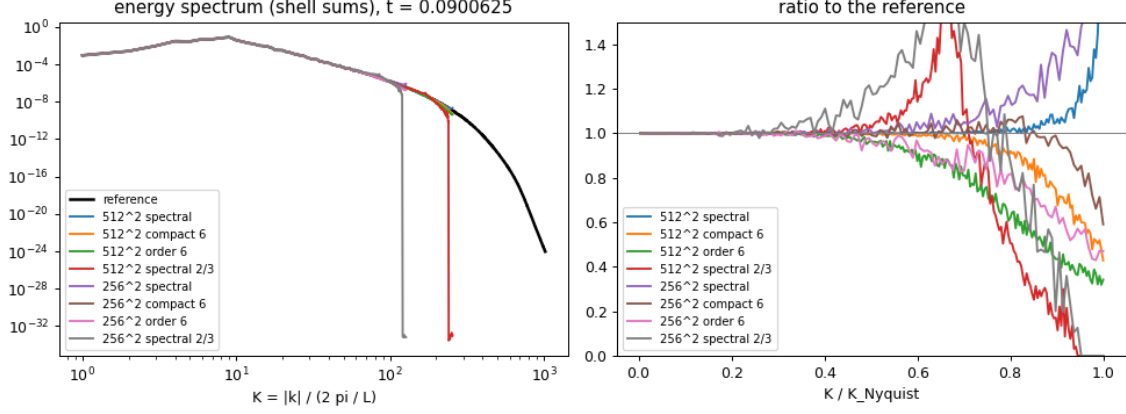


Figure 2: Baseline ($Re = 2 \times 10^4$, $k_0 = 10$) at $t = 0.09$: energy spectra on 256^2 and 512^2 and their ratio to the 2048^2 reference. The finite differences fall below the reference near the cutoff; undealiased pseudospectral piles energy up there (twice the reference at the Nyquist shell on 256^2); the 2/3 rule truncates and piles up just below its cutoff.

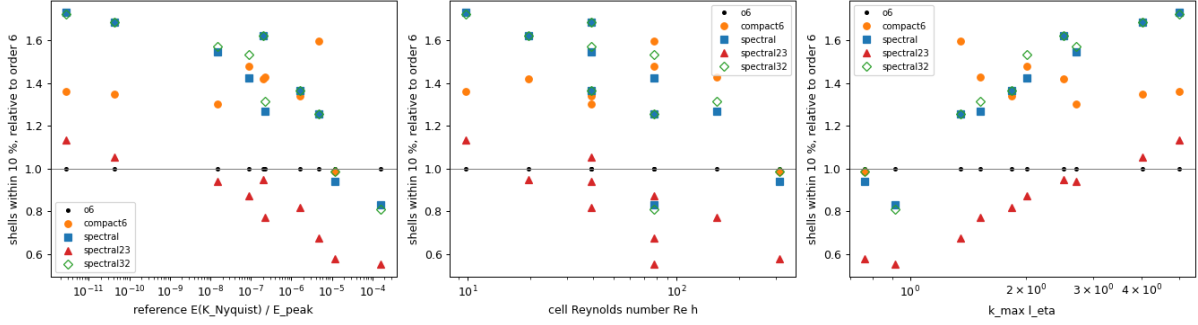


Figure 3: Resolved bandwidth relative to E6 for the five decaying cases on two grids, against χ (left), the cell Reynolds number $Re_c = Re h$ (middle) and $k_{\max} l_\eta$ (right). Open diamonds: $S_{3/2}$, on top of the S squares. At $Re_c = 39$ and 78 the cases span the regimes; against $k_{\max} l_\eta$ the ordering changes monotonically.

- *Under-resolved*, $k_{\max} l_\eta \lesssim 1$ (0.76, 0.92): no high-resolution scheme beats E6. Its larger differentiation error near the cutoff acts there as a numerical attenuation. This does not make the E6 calculation resolved: it only gives the smallest spectral-bandwidth error among the schemes tested on a grid that is under-resolved for all of them, and does not restore the missing scales.
- *Resolved*, $1 \lesssim k_{\max} l_\eta \lesssim 2$ (1.35, 1.52): the compact scheme leads clearly, with 1.4–1.6 times the shells of E6 against 1.3 for pseudospectral differentiation, which falls behind, dealiased or not, because energy piles up at the cutoff (fig. 2; section 5.2). The finite differences underestimate the derivatives near the cutoff instead ($k^* < k$), and their spectra fall below the reference.
- *Crossover*, $k_{\max} l_\eta \approx 2$ (1.83, 2.01): compact and pseudospectral differentiation are within 4% of each other, in either order; padding makes the difference at 2.01.
- *Well resolved*, $k_{\max} l_\eta \gtrsim 2.5$ (2.50–5.01): pseudospectral differentiation resolves the most, 1.55–1.73 times the shells of E6 and 92–98% of the Nyquist range, against 1.30–1.42 for compact. The margin over compact, 15–53 shells, far exceeds the agreement of the two references (1.5×10^{-3}).

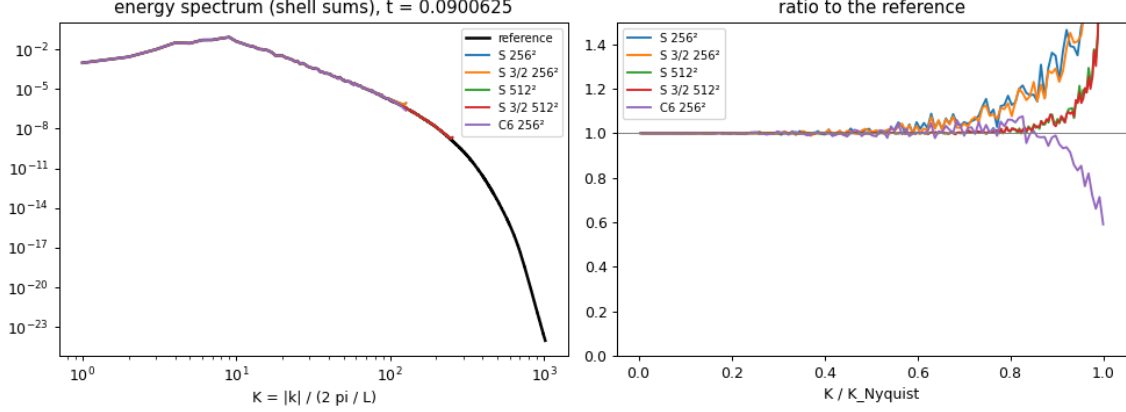


Figure 4: Baseline at $t = 0.09$: pseudospectral differentiation with and without 3/2 padding on 256^2 and 512^2 , and C6 on 256^2 , against the 2048^2 reference (right: ratio, clipped at 1.5). The padded and unpadded curves lie on top of each other.

Wherever $k_{\max} l_{\eta} \gtrsim 1$, the compact scheme resolves 1.3–1.6 times the shells of E6. The 2/3 rule caps the range at 2/3 of the Nyquist wavenumber by construction, and its spectrum piles up just below its own cutoff; it never leads.

The boundaries fall near the thresholds quoted for a resolved ($k_{\max} l_{\eta} > 1$) and a well-resolved (≥ 2) two-dimensional simulation [24]; with their $k_{\max}$ they would be 6% lower. By the bandwidth and cost measures used here, explicit sixth order does best on an under-resolved grid (which no scheme makes adequate; the remedy for that is resolution), compact sixth order on a resolved but not well-resolved one, and pseudospectral differentiation on a well-resolved one. This is an organization of these ten cases, found and shown on the same data, not a validated prediction. Ten points place the boundaries only roughly, between 0.92 and 1.35 and between 1.5 and 2.5. χ , a cruder proxy for the same thing, orders the cases approximately but not monotonically: two pairs in the crossover band (2.0 and 2.3×10^{-7} ; 9.0×10^{-8} and 1.6×10^{-6}) come out in the wrong order.

5.2 Is the pile-up aliasing? The 3/2 control

The natural explanation of the pseudospectral pile-up is aliasing: with no numerical attenuation near the cutoff, products of resolved modes alias back onto the highest retained ones [7, 8]. $S_{3/2}$ tests it. It removes the quadratic aliasing of the nonlinear term from every mode but the Nyquist ones and keeps the full range, which $S_{2/3}$ does not.

- *Observed.* In all ten cases $S_{3/2}$ resolves the same bandwidth as S, within 8 shells (table 2; the open diamonds of fig. 3). The pile-up is the same: the largest E/E_{ref} is 1.6–2.5 with padding and without, and the two spectra are indistinguishable near the cutoff (fig. 4). Padding changes the leader in one case ($k_0 = 5$ on 256^2 , 109 shells against C6’s 105), and moves no regime boundary.
- *Established.* With the skew-symmetric form used throughout, quadratic aliasing of the nonlinear term is not what causes the pile-up, nor the loss of pseudospectral bandwidth relative to C6 in the resolved regime. The skew-symmetric form is known to reduce aliasing errors [38, 39]; whether the conclusion holds for the advective form was not tested. The crossover between compact and pseudospectral differentiation survives 3/2 dealiasing of the retained non-Nyquist modes.
- *Suggested.* The data are consistent with an accumulation caused by the truncation itself: the enstrophy the cascade carries to the cutoff is not dissipated fast enough before it, and

ms per GPU step	E6 (sparse)	E6, C6, S (Fourier)	S _{2/3}	S _{3/2}	C6, banded sparse
256 ²	3.6	4.8	5.0	7.7	66
512 ²	12.8	17.7	18.4	31.5	269
1024 ²	51.8	78.5	82.0	–	–

Table 3: Time per step (decaying turbulence, skew-symmetric form, RK4). Applied in Fourier space, all schemes cost the same; seven-point stencils as sparse products are cheaper still. The compact operator as an exact banded circulant (78 and 45 entries per row) costs 15 times more than in Fourier space. Padding costs 1.85 times the unpadded step: five transforms on $(3n/2)^2$ points per evaluation of the nonlinear term.

collects in the last retained shells, as in Galerkin-truncated flows [28, 16, 29]. This reading is not new: it has been stated for alias-free three-dimensional simulations [31], and the 2/3 method is known to share the high-mode pathology of the aliased one once the solution is not smooth [30]. What we found no precedent for is the control on one marginally resolved flow. On this reading the finite differences’ $k^* < k$ near the cutoff attenuates the transfer into the last shells, and that is why their spectra fall below the reference instead.

- *Not established.* The mechanism itself. These runs rule out one explanation; testing the truncation reading, for instance with a selective filter near the cutoff, is a separate study.

5.3 Cost at equal resolved range

In Fourier space every scheme costs the same per step, about 1.4 times E6 applied as a sparse stencil (table 3). The schemes differ in their stable time step, by $(k^*h)_{\max}$, and in the range they resolve. At equal resolved range relative to E6, cost follows $k_{\max}l_{\eta}$ much as the bandwidth does.

- Below $k_{\max}l_{\eta} \approx 1$ E6 is the cheapest; C6 costs 1.9–3.1 times as much, S 3.4–4.9 times.
- From about 1 up to 2.5, C6 is the cheapest, at 0.44–0.74 times E6; S costs 0.64–1.40 times.
- Above about 2.5, S is the cheapest, at 0.54–0.76 times E6; C6 costs 0.69–0.81 times.
- S_{2/3} never costs less (1.3–11 times).
- Nor does S_{3/2}. It resolves what S resolves at 1.85 times the step, so it costs 1.00–1.31 times E6 for $k_{\max}l_{\eta} \geq 2$ and 2–9 times below. (Both step times from the same build: 31.5 against 17.0 ms on 512².) At equal resolved range it is still cheaper than S_{2/3} in every case, as Sinhababu and Ayyalasomayajula [35] found for under-resolved flows.

The gains of the high-resolution schemes are modest, tens of per cent, where they exist. They come from resolving more shells, against a time step that is 20 % (C6) to 50 % (S) smaller than E6’s. These time-step ratios hold for the skew-symmetric form; with an Arakawa Jacobian, spectral-like differentiation can tighten the limit considerably [34]. Times are from the build used for all comparisons (the baseline rerun with it).

5.4 Forced turbulence

In forced turbulence the comparison is between time-mean statistics. With 28 forced modes, the realized enstrophy input over the averaging windows varies by about $\pm 8\%$, which sets a floor below which the schemes cannot be distinguished.

With hyperviscosity, all four schemes give the same spectral slope in the direct cascade (−3.44 to −3.48). The enstrophy flux leaves its plateau at the same shell, $K = 66$, where the hyperviscosity takes over. The plateau levels (41.9, 43.0, 46.2, 50.1 for E6, C6, S, S_{2/3}) differ as

much as the total enstrophy each run removed, so they reflect the realized input, not the cascade. When the dissipation is confined near the cutoff, the differentiation scheme makes no measurable difference to the inertial range.

With plain viscosity ($\text{Re} = 10^5$; at the 256^2 cutoff $k_{\text{max}}l_\eta = 1.33$, from the reference’s time-mean enstrophy dissipation, and $\chi = 2.8 \times 10^{-7}$: the band where compact leads the decaying cases), each spectrum is normalised by its own enstrophy flux, $E \propto \Pi_Z^{2/3}$, to remove the realized input. The spectra then agree with the 768^2 reference to 10 % up to $K = 100$ (E6), 89 (S), 84 (C6) and 42 ($S_{2/3}$). E6 has the largest nominal K_{10} , unlike the decaying cases at the same $k_{\text{max}}l_\eta$, but the differences are within the remaining scatter: away from the cutoff the normalised spectra still differ from the reference by up to 8 %. What stands out is how each scheme behaves at the cutoff. At $K = 120$, E6 falls below the reference (0.64) and S piles energy up (1.53, and 2.0 at the Nyquist shell). C6 lies in between (1.00, with an overshoot of up to 17 % just below), and $S_{2/3}$ truncates. This agrees with the finding of Rodhiya et al. [10] that forced turbulence computed with spectral and finite-difference methods has nearly the same statistics.

6 Discussion

The results answer the three questions of the introduction for the cases tested. First, the axis. San and Staples [9] organized resolution by Re_c , a property of the grid and the viscosity. At fixed Re_c we find every regime: the flow’s realized dissipation scale, which depends on its spectrum and not only on the viscosity, decides. The classical measure of it, $k_{\text{max}}l_\eta$, orders the ranking monotonically, with boundaries near the thresholds long used to call a two-dimensional simulation resolved and well resolved [24]. The practical reading is that familiar criterion, used to choose the scheme rather than only to trust a run. Here it is an a posteriori coordinate: l_η came from the reference. An a priori estimate of the enstrophy dissipation would be needed to choose a scheme before running, and an under-resolved run may misjudge it; that was not tested. Their finding that sixth-order schemes agree with the spectral method when well resolved, like that of Browning and Kreiss [21], does not contradict ours: the resolved bandwidth is a stricter measure near the cutoff, and even at large $k_{\text{max}}l_\eta$ pseudospectral differentiation resolves up to 1.7 times the shells of E6. Their speed-ups of about 5 compared finite differences with a dealiased spectral method whose cost was interpolated. Here one code applies every scheme, E6 as a sparse stencil and the others in Fourier space; C6 and S cost 1.4 times E6 per step, padding 1.85 times S, and the differences in cost at equal range come from these, the time step and the resolved range.

Second, the pile-up. San and Staples saw pile-up in non-conservative finite differences and attributed it to aliasing. Here the nonlinear term is skew-symmetric and conserves enstrophy for every scheme, the finite differences never pile up, and the pile-up of pseudospectral differentiation survives $3/2$ padding. With this form, quadratic aliasing of the nonlinear term is therefore not the cause. That is consistent with the truncation reading of the literature [31, 30, 16], and with the dispersion-driven pile-up of Yalla et al. [15], but these runs do not identify the mechanism.

Third, the scheme. The results place the quasi-spectral argument of Laizet and Lamballais [2] on a map. Compact sixth-order differentiation is the most efficient choice where the grid resolves the enstrophy-dissipation scale but not well ($1 \lesssim k_{\text{max}}l_\eta \lesssim 2$): marginal but not overwhelmed, the regime they targeted. When the grid resolves it well, the exact symbols of pseudospectral differentiation give more bandwidth for the cost. When it does not resolve it, explicit sixth order is the most robust: its underestimated derivatives near the cutoff act as attenuation, as the lower-order schemes of Vreman et al. [22] did under coarse filtering. That is the least bad of the under-resolved calculations, not a substitute for resolution. Both ways of dealiasing gain exact conservation; the $2/3$ rule pays for it with a third of the range, padding with 1.85 times the step, and in our measures neither pays off. Cheaper dealiasing by phase shifts [36] or high-order Fourier filtering [37], or a selective filter [17], could change the cost of the spectral variants and were not tested.

Two qualifications matter. First, the map organizes transient decaying turbulence before decorrelation, not all turbulence statistics. The forced viscous case sits at $k_{\max}l_\eta = 1.33$, where compact leads the decaying cases; there explicit sixth order has the largest nominal K_{10} , but the differences are within the remaining sampling scatter, and the inertial range does not distinguish the schemes. Second, the comparison measures accuracy as spectral bandwidth; other quantities, such as vortex statistics, higher-order moments or structure functions, may weigh the cutoff differently.

The apparatus has a methodological point of its own. Applying finite-difference operators through their symbols in Fourier space is natural for periodic grids, and is related to the mixed physical/spectral treatment of [2]. Here it is used as an experimental control: the scheme is a table of symbols, and the solver, the Poisson operator and the diagnostics derive from the same table. This rules out, for instance, a Poisson operator and a transport Laplacian that disagree. It also made the compact scheme 15 times faster than its exact banded form, which the comparison of cost needed. On non-periodic or distributed grids the compact scheme needs tridiagonal solves instead, whose cost on GPUs is a subject of its own [18].

Limitations. The study is two-dimensional and periodic. It uses one time integrator (RK4), one nonlinear form (skew-symmetric) and one GPU in double precision, on which fast Fourier transforms dominate the step time. The matrix of cases is small: four perturbations of one decaying baseline and two forced cases on 256^2 , which samples $k_{\max}l_\eta$ at ten values and places the regime boundaries only roughly. Selective filtering near the cutoff, which Yamamoto and Tsuji [17] found the most cost-effective in channel flow, was not tested. The forced comparisons are limited by the sampling scatter of the random forcing. Wall-bounded flows, where compact schemes need boundary closures and spectral methods lose their simplicity, are outside its scope.

7 Conclusions

For the tested two-dimensional periodic turbulence, no differentiation scheme is best everywhere. In decaying turbulence the ranking is not organized by the cell Reynolds number, since cases at equal Re_c fall in every regime, but by how well the grid resolves the flow’s enstrophy-dissipation scale, measured a posteriori. Explicit sixth order resolves the most and costs the least when $k_{\max}l_\eta \lesssim 1$ (the least bad of the under-resolved calculations, not a resolved one), compact sixth order when $1 \lesssim k_{\max}l_\eta \lesssim 2$, and pseudospectral differentiation when $k_{\max}l_\eta \gtrsim 2$ –2.5: the changes in leading scheme occur near the thresholds quoted for a resolved and a well-resolved simulation. The ordering of the cases does not depend on the wavenumber convention, which rescales every $k_{\max}l_\eta$ alike; the correspondence with those thresholds does. The crossover between compact and pseudospectral differentiation survives 3/2 dealiasing of the retained non-Nyquist modes, which leaves the pseudospectral pile-up and bandwidth unchanged at 1.85 times the step cost; with the skew-symmetric form it is not an artifact of quadratic aliasing. The organization is that of transient decaying turbulence: in a forced, statistically stationary case at $k_{\max}l_\eta = 1.33$ the schemes are indistinguishable in the inertial range within the forcing’s sampling scatter, differing only at the cutoff. Open are an a priori estimate of l_η that would make the rule predictive, more points to place the boundaries, the advective form, and three dimensions.

Code and data availability. The solver, the analysis tools (`tools/compare_schemes.py`, `tools/cascade.py`, `tools/modified_wavenumber.py`) and the command lines of every run are in the cnavier repository (<https://github.com/leo-aa88/cnavier>), with a detailed methodology document (`docs/cnavier_methodology.pdf`).

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